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Capstone Project

Smart Agriculture Farm Management System

CSA0211 – C Programming

Guided by

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Problem Statement

- Traditional farm record management is mostly manual, making it difficult for farmers to organize crop information, monitor resources, track expenses, and maintain harvest records.
- Paper-based documentation often results in data loss, calculation errors, and inefficient decision-making.
- There is a need for a simple software system that digitally stores farm information using C programming concepts and file handling, enabling efficient record management and improved agricultural productivity.



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ABSTRACT

The Smart Agriculture Farm Management System is a console-based application developed in C to simplify agricultural record management.

The system enables farmers to:

- Register crop details
- Manage cultivation activities
- Track farm resources and expenses
- Record harvest information
- Store data permanently using files

The application uses arrays, strings, functions, and file handling to organize agricultural records. It minimizes paperwork, improves accuracy, and provides an efficient method for maintaining farm information.



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INTRODUCTION

- Agriculture requires efficient management of crop cultivation, farm resources, expenses, and harvest records.
- Traditional paper-based record keeping is time-consuming, prone to errors, and difficult to maintain.
- The **Smart Agriculture Farm Management System** is a console-based application developed using the **C programming language**.
- It enables farmers to digitally manage crop records, resource usage, expenses, and harvest information.
- The system uses **Arrays, Strings, Functions, Structures, and File Handling** to store and process farm data efficiently. It provides a simple, low-cost, and reliable solution for improving farm management and productivity.





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Existing System

- Paper-based record maintenance
- Manual crop registration
- Separate notebooks for expenses
- Difficult to calculate total production
- High chance of data loss
- Time-consuming record searching
- No centralized farm management

Limitations

- Human errors
- Duplicate records
- Poor data security
- Difficult reporting
- No automated calculations



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Proposed System

The proposed Farm Management System provides a digital platform for maintaining agricultural records.

Benefits

- Easy record management
- Reduced paperwork
- Accurate calculations
- Fast information retrieval
- Better decision making
- Low-cost software

Features

- Crop Registration
- Cultivation Record Management
- Resource Tracking
- Expense Management
- Harvest Recording
- Production Summary
- File-based permanent storage
- Search, Update and Delete records



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Literature Survey

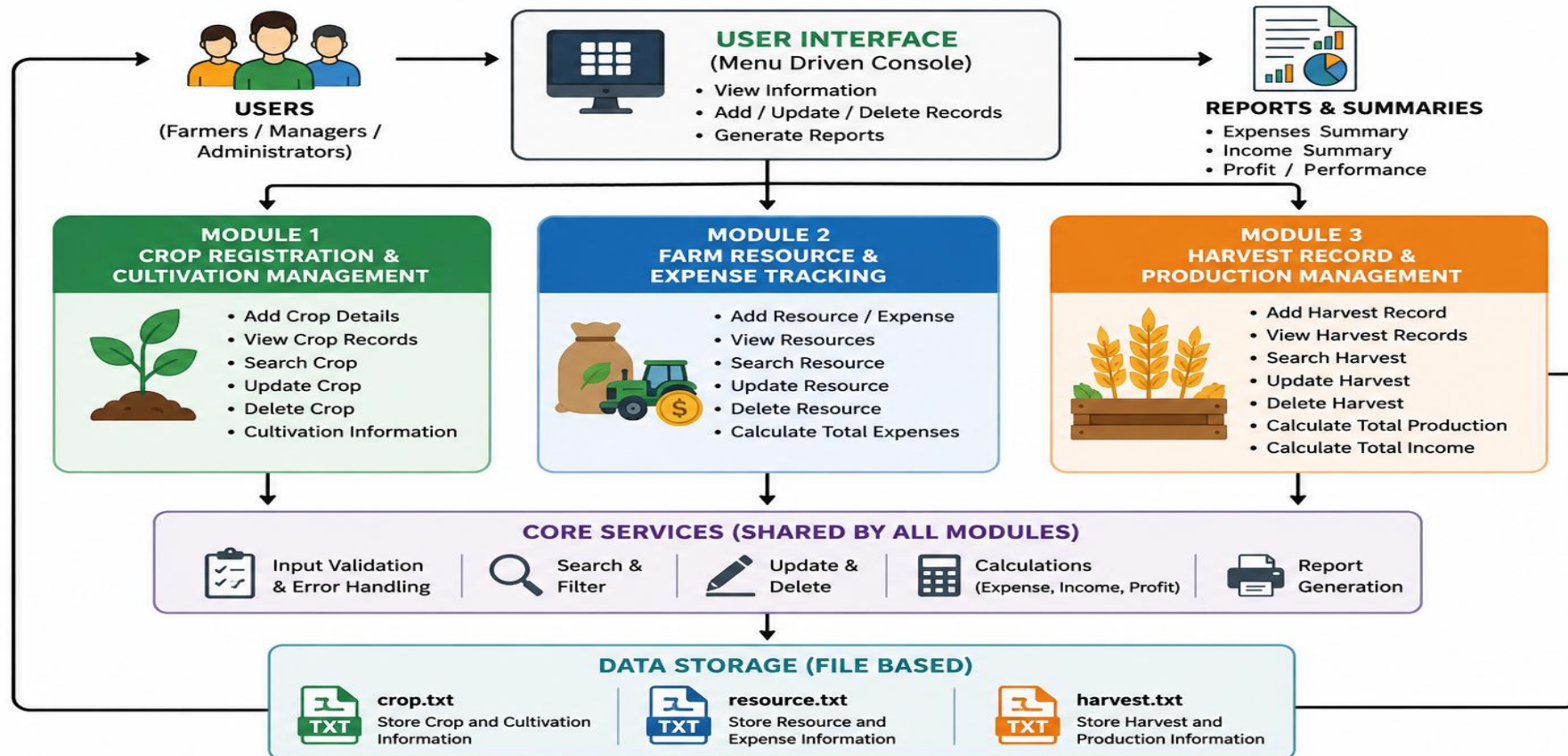
Author & Year	Study / Focus	Key Findings	Relevance to Our Project
Tummers et al. (2021)	Farm Management Information Systems	Identified crop, field and financial management as key functions.	Supports centralized farm record management.
Iaksch et al. (2021)	Digitalization & Big Data in Smart Farming	Digital data improves farm management and decision-making.	Supports digital storage and processing of farm data.
El Idrissi et al. (2022)	ICT in Intelligent Farming	Discussed digital technologies for improving agricultural management.	Provides background for smart agriculture systems.
Garg & Alam (2023)	Smart Agriculture Technologies	Highlighted IoT, cloud, AI and data analytics in modern farming.	Shows future scope for extending our system.
Basir et al. (2024)	Digital Farm Recordkeeping	Discussed migration from paper-based to digital farm records.	Directly supports replacing manual farm records.
Indian Digital Agriculture Study (2023)	Farm Management Applications	Reviewed digital agricultural applications and adoption challenges.	Highlights the need for simple and accessible solutions.

Research Gap

Existing research largely focuses on advanced technologies such as IoT, AI, cloud computing and big-data analytics. Our project focuses on a simple, low-cost farm management system using core C programming concepts and file-based data storage.

Smart Agriculture Farm Management System

THREE MODULE BASED SYSTEM ARCHITECTURE





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Module 1 – Crop Registration & Cultivation Management

Objective

To maintain complete records of crops cultivated on the farm and monitor their cultivation details.

Features

- Register new crop information
- Record planting and expected harvest dates
- Store crop type, season, and cultivated area
- Search, update, and delete crop records
- View all registered crops
- Save data permanently using file handling

Output

- Organized crop database
- Easy access to cultivation records
- Improved crop planning and monitoring

```
=====
Enter your choice (1-5): 1
=====
Crop Registration
=====
1. Add Crop
2. View All Crops
3. Search Crop
4. Update Crop
5. Delete Crop
6. Back to Main Menu
=====
Enter choice: 2
=====
All Crops
=====
ID | Name | Type | Season | Area | Farmer
-----
101 | Wheat | Cereal | Winter | 50.50 | John Smith
102 | Corn | Grain | Summer | 75.00 | David Miller
103 | Soybeans | Legume | Summer | 40.00 | Sarah Jones
104 | Tomatoes | Vegetable | Spring | 15.50 | Carlos Ruiz
105 | Cotton | Fiber | Summer | 60.00 | Emma White
1 | Wheat | Summer | summer | 3.00 | vijay
```

Crop Registration & Cultivation Management in Terminal



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Module 2 – Farm Resource & Expense Tracking

Objective

To monitor farm resources and manage operational expenses efficiently.

Features

- Record seed, fertilizer, pesticide, and irrigation usage
- Track labor and machinery expenses
- Update and delete resource records
- Calculate total farming expenses automatically
- Generate expense summary reports
- Store all records in files

Output

- Accurate expense management
- Better resource utilization
- Reduced financial tracking errors

```
=====
Farm Resource & Expense Tracking
=====
1. Add Resource
2. View All Resources
3. Update Resource
4. Delete Resource
5. Back to Main Menu
=====
Enter choice: 2

=====
All Resources
=====
```

ID	Type	Name	Quantity	Total Cost
201	Seed	Wheat Seeds (Premium)	500.00 kg	\$600.00
202	Fertilizer	Nitrogen NPK 10-10-10	1200.00 kg	\$1020.00
203	Chemical	Pesticide (Organic)	50.00 L	\$750.00
204	Fuel	Diesel for Tractors	1000.00 L	\$1100.00
205	Labor	Seasonal Workers	500.00 hours	\$6250.00
206	Equipment	Irrigation Pipes	200.00 m	\$1100.00

Farm Resource & Expense Tracking in Terminal



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Module 3 – Harvest Record & Production Management

Objective

To maintain harvest details and analyze farm production.

Features

- Record harvest information for each crop
- Store harvest date and quantity produced
- Record selling price and calculate total income
- Search, update, and delete harvest records
- Generate production and income reports
- Maintain permanent harvest records using file handling

Output

- Accurate production records
- Automated income calculation
- Better analysis of farm productivity and profitability

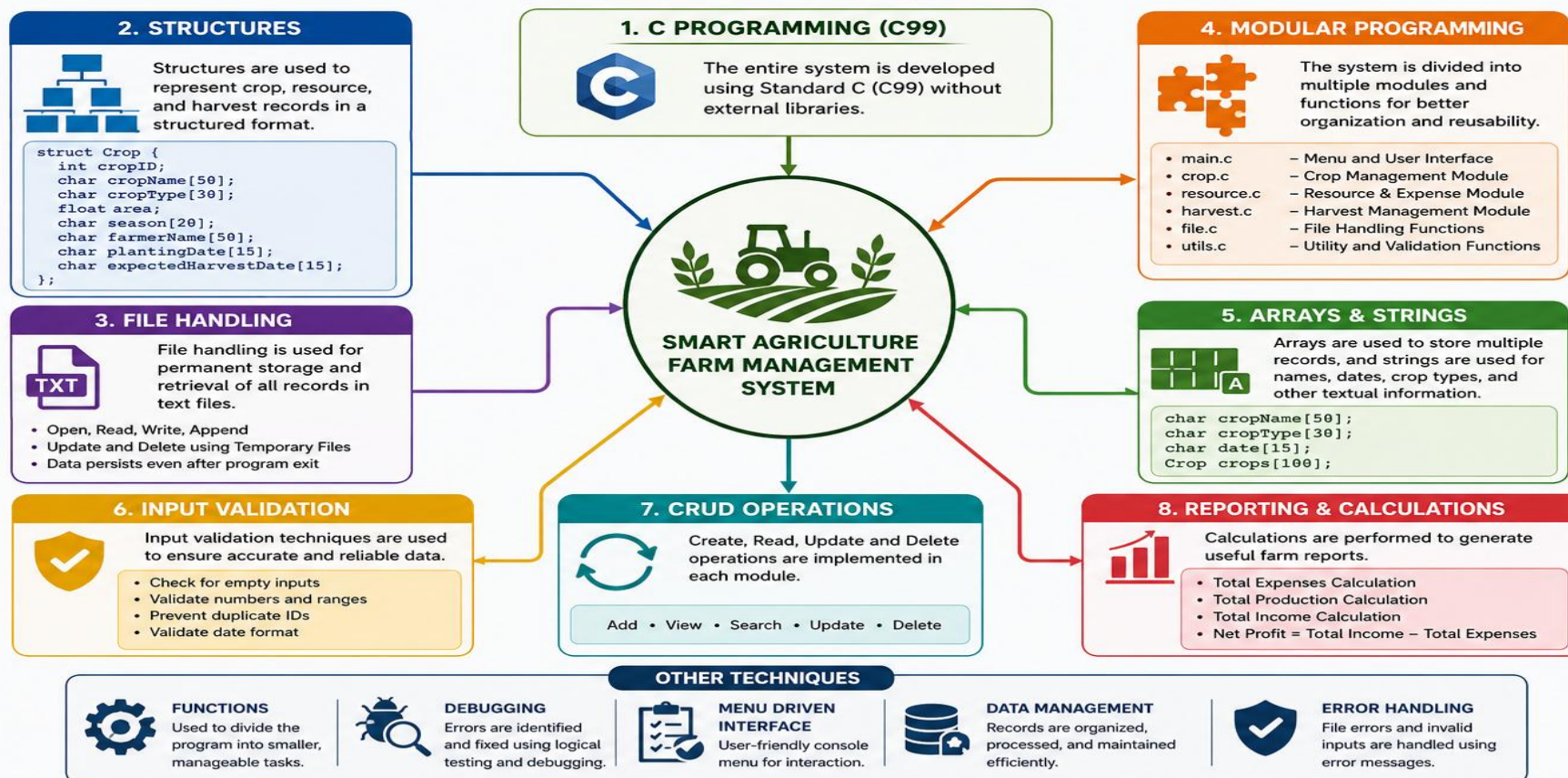
```
=====
Harvest & Production Management
=====
1. Add Harvest Record
2. View All Harvests
3. Delete Harvest Record
4. Back to Main Menu
=====
Enter choice: 2

=====
All Harvest Records
=====
ID | Crop ID | Date | Quantity | Income
-----
301 | 101 | 2026-04-15 | 150000.00 kg | $52500.00
302 | 104 | 2026-08-05 | 25000.00 kg | $45000.00
303 | 102 | 2025-09-20 | 300000.00 kg | $84000.00
```

Harvest Record & Production Management in Terminal

TECHNIQUES USED

Smart Agriculture Farm Management System





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RESULTS AND DISCUSSION

Results

- Successfully developed a menu-driven farm management system.
- Crop, resource, and harvest records are stored using file handling.
- Supports Add, Search, Update, Delete, and View operations.
- Automatic calculation of expenses and harvest income.

Discussion

- Reduces manual record-keeping errors.
- Improves data organization and retrieval.



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CONCLUSION

- The Smart Agriculture Farm Management System provides a simple and efficient solution for managing farm records digitally. It enables users to maintain crop, resource, and harvest information with greater accuracy while reducing manual effort. By implementing arrays, strings, functions, and file handling, the project demonstrates the practical application of core C programming concepts. The system can be further enhanced by integrating databases, IoT devices, or mobile applications for advanced farm management.



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THANK YOU